

Introduction to some probability distributions

Using python.

Iván Andrés Trujillo

ivantrujillo1229

ai-page.readthedocs.io

What distribution we are modelled?

- Binomial
- Poisson
- Exponential

Bernoulli Distribution

In a single trial with only two possible outcomes; *Success*(1) or *Failure*(0).

Roll a dice

Assume that you win if lands $\{3, 4, 5, 6\}$ or lost if lands $\{1, 2\}$. The probability of win is $\frac{4}{6}$ and lost is $\frac{2}{6}$.

Probability Mass Function (PMF)

X	0	1
$f(X = x)$	$(1 - P)$	P

$$P(X = x) = P^x(1 - P)^{1-x} \quad (1)$$

Mean and variance

Mean

$$\mu = E(X) = \sum f(x)x = P \quad (2)$$

Variance

$$\begin{aligned} \sigma^2 = \text{Var}(X) &= E(X^2) - (E[X])^2 \\ &= P - P^2 \\ &= P(1 - P) \end{aligned} \quad (3)$$

Flipping a coin

- Random variable x will take the value 1 (success) when the coin land Tail.
- Each trial is independent
- The probability is constant

Problem

Determine the probability of get three successes ($k = 3$) in five trials ($n = 5$).

5 trials and 3 success...

5 trials and 3 success...

Success and failures	Probability
$E_1 E_2 E_3 F_4 F_5$	$p^3(1-p)^2$
$E_1 E_2 E_4 F_3 F_5$	$p^3(1-p)^2$
$E_1 E_2 E_5 F_3 F_4$	$p^3(1-p)^2$
$E_1 E_3 E_4 F_2 F_5$	$p^3(1-p)^2$
$E_1 E_3 E_5 F_2 F_4$	$p^3(1-p)^2$
$E_1 E_4 E_5 F_2 F_3$	$p^3(1-p)^2$
$E_2 E_3 E_4 F_1 F_5$	$p^3(1-p)^2$
$E_2 E_3 E_5 F_1 F_4$	$p^3(1-p)^2$
$E_2 E_4 E_5 F_1 F_3$	$p^3(1-p)^2$
$E_3 E_4 E_5 F_1 F_2$	$p^3(1-p)^2$

Binomial distribution

We must said that $X \sim B(k, n, p)$

Probability Mass Function (PMF)

$$P(X = x) = \binom{n}{x} P^x (1 - P)^{n-x} \quad (4)$$

Cumulative Distribution Function (CDF)

$$P(X \leq x) = \sum_{i=0}^x \binom{n}{i} P^i (1 - P)^{n-i} \quad (5)$$

```
from scipy.stats import binom
binom.pmf(succeses, trials, P)
binom.cdf(succeses, trials, P)
```

Test extreme cases:

- $CDF(n, n, 0.5)$
- $PMF(1, 1, 0.5)$

Mean and variance sum of independent Bernoulli trials

Let $B_1, \dots, B_n$ iid random variables with $\mu = P$ and $\sigma^2 = P(1 - P)$ if think binomial as the sum of this variables:

Mean of binomial distribution

$$E(X) = \sum_{i=1}^n E(B_i) = nP. \quad (6)$$

Variance of binomial distribution

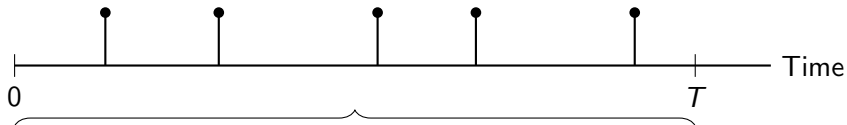
$$\text{Var}(X) = \sum_{i=1}^n \text{var}(B_i) = nP(1 - P) \quad (7)$$

In what we are interested?

No negative events

- Number calls in a call center
- Number of arrivals in a gas stations
- Visits a web site
- Number of traffics accidents per year, etc...
- Number of disease by a disease.

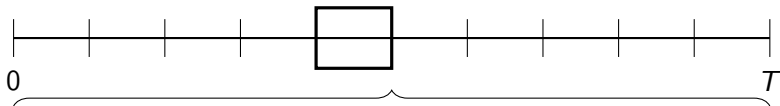
Arrivals during the interval



Expected number of arrivals: λ

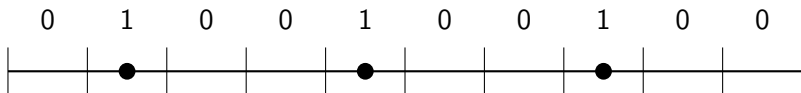
$$P(\text{arrival in slot}) = p$$

One small slot



n equal slots

$$p = \frac{\lambda}{n}$$



$$X_i = \begin{cases} 1 & \text{arrival} \\ 0 & \text{no arrival} \end{cases}$$

$$X = X_1 + \cdots + X_n$$

$$X \sim \text{Binomial}(n, p)$$

Poisson distribution

Binomial distribution tends to poisson as $n \rightarrow \infty$, $p \rightarrow 0$, $\lambda = np$ is constant.

remember that the expected value of a binomial is np .

remember that binomial is

$$\binom{n}{x} p^x (1-p)^{n-x} \quad (8)$$

Remember that the expected value of occurrences in binomial is np fixed!.

assuming that $\lambda = np$ (why?) think as $n \rightarrow \infty$ a lot of trials! then $p = \frac{\lambda}{n}$, then this have a lot of sense.

$$\begin{aligned}\binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} &= \frac{n!}{x!(n-x)!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\&= \frac{\lambda^x}{x!} \frac{n!}{(n-x)!n^x} \left(1 - \frac{\lambda}{n}\right)^{n-x} \\&= \frac{\lambda^x}{x!} \frac{n(n-1)\dots(n-x+1)}{n^x} \left(1 - \frac{\lambda}{n}\right)^{n-x} \\&= \frac{\lambda^x}{x!} \frac{n}{n} \frac{(n-1)}{n} \dots \frac{(n-x+1)}{n} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x}\end{aligned}$$

now applying the limit we have:

$$\lim_{n \rightarrow \infty} \left[\frac{\lambda^x}{x!} \frac{n}{n} \frac{(n-1)}{n} \dots \frac{(n-x+1)}{n} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x} \right]$$

applying limit property the limit of products is the product of limits we have and the limit of a constant is the constant.

$$\frac{\lambda^x}{x!} \left(1 - \frac{1}{n}\right) \dots \left(1 - \frac{x-1}{n}\right) \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x} = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!} \quad (9)$$

Have a lot of applications. x means the number of events that do occur in the unit of time.

...

A bank receives an average of three credit card fraud claims per hour.

- What is the probability of receiving exactly 5 claims in one hour?
- What is the probability of receiving at least 2 claims in one hour?

...

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```
from scipy.stats import poisson
LAMBDA = 3
print(poisson.pmf(5,LAMBDA))
print(1 - poisson.cdf(1, LAMBDA))
```

Examples

...

In a fintech company, loan applications arrive at an average rate of 14 per hour.

- What is the probability of receiving exactly 100 loan applications in one 8-hour working day?
- What is the probability of receiving no loan applications in one hour?

Examples

...

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```
from scipy.stats import poisson
LAMBDA = 14
LAMBDA_DAY = LAMBDA * 8
print(poisson.pmf(100, mu=LAMBDA_DAY))
print(poisson.pmf(0, mu=LAMBDA))
```

- [Lecture 1 - \(click here\)](#)
- [Lecture 2 - \(click here\)](#)
- [Lecture 3 - \(click here\)](#)

...

[Bombs simulation \(click here\)](#)

Horse kick data

Now we are present horse kick data, we can replicate the results?
page 25 of Bortkiewicz's Law of Small Numbers.
name table Deaths by horse kick in the Prussian army 1894 - 1875.

Deaths by year	Cases occurred	Expected number
0	109	?
1	65	?
2	22	?
3	3	?
4	1	?
5 and more	-	?

...

Poisson distribution could be predict those results? We can validate?, yes using MLE!

Lab

[click here](#)

Deaths by year	Cases occurred	Expected number
0	109	108.7
1	65	66.3
2	22	20.2
3	3	4.1
4	1	0.6
5 and more	-	0.1

The power of statistics!

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Es importante entender lo siguiente

Web visitis

We need model the visit in time in our we page, we can uses poisson.

Poisson lab

ver laboratorio aquí (**click here**)

$$P(X_i = x_i) = \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \quad (10)$$

by independence:

$$L(\lambda) = \prod_i^n P(X_i = x_i \mid \lambda) \quad (11)$$

$$\ell(\lambda) = \log L(\lambda) \quad (12)$$

$$\ell(\lambda) = -n\lambda + \left(\sum x_i\right) \log \lambda - \sum \log(x_i!) \quad (13)$$

$$\frac{d\ell(\lambda)}{d\lambda} = -n + \frac{\sum x_i}{\lambda} \quad (14)$$

to take the maximum:

$$\lambda_{MLE} = \frac{1}{n} \sum x_i \quad (15)$$

...

The maximum likelihood estimator of the poisson parameter is the sample mean!

Important

λ is the expected number of events per unit time. $\lambda = 3$ arrives three patients by hour. if we need the probability at a longer interval, the expected number is proportional with time λt .

Find the probability in two hours

We need find out the probability of arrives three patients in two hours. $\lambda 2 = (3)2 = 6$.

more generally we can write the poisson distribution as:

$$\frac{(\lambda t)^x}{x!} e^{-(\lambda t)} \quad (16)$$

Also we can write $\mu = \lambda t$ as the number of arrivals in that interval λ rate(events per unit time).

Poisson process

What is a poisson process?

Events that occur one by one in a given in time, randomly and independently (that occur in one interval not affect others), where occur in average (Constantly) λ events by unit of time.

Examples

- calls to a callcenter
- arrivals to a keepstore
- etc..

Exponential distribution ideas: The probability of get 0 occurrences in the interval t in a Poisson process is

$$P(x = 0) = e^{-\lambda t} \quad (17)$$

...

How many units of time we need wait until occur the first event?

Assume that we have an **poisson process** with a positive λ rate.

$$P(N(t) = x) = \frac{(\lambda t)^x e^{-\lambda t}}{x!}, \quad k = 0, 1, \dots \quad (18)$$

Note, x is the number of arrivals in the interval $[0, t]$. Be T the time **until** occur the first event, the probability that not occur an event in the interval $[0, t]$ is:

$$e^{-\lambda t} \quad (19)$$

The event $\{T > t\}$ means that the first occur **after** the time t is equally that not occur an event in the interval $[0, t]$ (there are not events).

$$P(T > t) = e^{-\lambda t} \quad (20)$$

at least one event occur in the interval $[0, t]$.

$$1 - P(N(t) = 0) \quad (21)$$

is equivalent to ask:

$$P(T \leq t) = 1 - e^{-\lambda t} \quad (22)$$

note que esto no es más que la *CDF*, para obtener la *pdf*

$$\frac{d(1 - e^{-\lambda t})}{dt} = \lambda e^{-\lambda t} \quad (23)$$

Applications of exponential distribution

To model the amount of waiting time until the following event.

- The amount of time until a new version of a package is released
- The amount of time until a patient out of hospital.
- and so on

Exponential how is modelled?

para modelar con python se usa $scale = \frac{1}{\lambda}$ where λ is the already known parameter **average number of events by unit of time**.

```
from scipy.stats import expon
LAMBDA = 1# clientes per hour
expon.cdf(x = 1, scale = 1/LAMBDA) # we are modelling the
    times
# The probability that a client arrives in lesser than
    one hour
```

Exponential estimator

$$L(\lambda) = \prod \lambda e^{-\lambda x_i} \quad (24)$$

$$\ell(\lambda) = \log L(\lambda) \quad (25)$$

$$\ell(\lambda) = n \log \lambda - \lambda \sum x_i \quad (26)$$

$$\ell'(\lambda) = \frac{n}{\lambda} - \sum x_i \quad (27)$$

FOC:

$$\lambda_{MEL} = \frac{1}{\bar{x}} \quad (28)$$

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- Which is the probability of a person must be wait more than 10 minutes until the bus arrive.

Exponential distribution

- **Poisson:** The number of orders that arrive during a fixed period of time.
- **Exponential:** The waiting time until the next order arrives.

...

Clients arrival randomly and independent with an average frequency then we can use exponential distribution to model the time to the arrival of another client.

$$P(T > s + t \mid T > s) = P(T > t) \quad (29)$$

...

The probability that wait t minutes given that are passed s not bearing the past in future!

widely used to model waiting times and event duration.

Bus paradox or inspection paradox

- Rolle theorem
- lagrange theorem
- Cauchy Theorem
- L'hospital theorem

closed interval $[a, b]$ include a, b open interval (a, b) not include a, b .

Extreme value theorem

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if f is continuous in a closed interval $[a, b]$ then f has both a maximum and minimum.

...

Also we can say $\exists c, d \in (a, b) : f(c) \leq f(x) \leq f(d) \forall x \in (a, b)$.

What is closed bounded interval what is the difference?

Teorema de Rolle: existe un punto en el cual la velocidad instantánea es igual a la velocidad promedio; conecta esto en economía!

Cauchy Theorem

given $f, g : [a, b] \rightarrow \mathbb{R}$ continuous on $[a, b]$ and differentiable in (a, b) with $g'(x) \neq 0, \forall x \in (a, b)$ then exist a point $c \in (a, b)$ such that:

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}. \quad (30)$$

interpretation: $f(x)$ distance at time x , $g(x)$ wasted fuel until time x . $f'(x)$ instant velocity (k/h) $g'(x)$ instant compustion (l/h).

$$\frac{f(b) - f(a)}{g(b) - g(a)} \quad (31)$$

rendimiento promedio en (a,b) , cuantos kilometros se consiguen en promedio por cada litro de gasolina!

$$\frac{f'(c)}{g'(c)} \quad (32)$$

rendimiento instantáneo en el punto c . cuantos kilometros por litros logras en ese instante.

el teorema garantiza que en algún momento el rendimiento instantaneo coincide con el rendimiento promedio de todo el viaje!

General Mean Value Theorem

de dos funciones f y g relaciona la razón de los incrementos con la razón de las derivadas.

Si algo cambia de un valor a otro en un cierto intervalo, entonces en algún punto intermedio la velocidad instantánea de cambio es igual a la velocidad promedio de todo el intervalo.

important to solve indeterminations.

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{g'(a)}{f'(a)} \quad (33)$$

this is the special case

- $f(a) = g(a) = 0$ results in $\frac{0}{0}$
- $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty$ results in $\frac{\infty}{\infty}$

derivatives must exist and $g'(x) \neq 0$ near to zero. to proof L'hospital you need the generalized value mean theorem.

The definition of E is very interesting

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \quad (34)$$

Where this come from?

if $u = \frac{n}{x}$ then we have that

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \quad (35)$$

if you take:

$$\left(1 + \frac{x}{n}\right)^n = \left(1 + \frac{1}{u}\right)^{ux} \quad (36)$$

if $n \rightarrow \infty$ then $u \rightarrow \infty$,

$$\lim_{u \rightarrow \infty} \left[\left(1 + \frac{1}{u}\right)^u \right]^x = e^x. \quad (37)$$

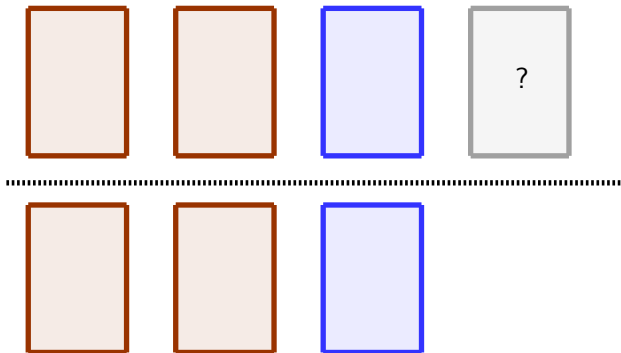
we know that

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n \quad (38)$$

assume that $g = -x$.

$$e^g = \lim_{n \rightarrow \infty} \left(1 + \frac{g}{n}\right)^n$$
$$e^{-x} = \lim_{n \rightarrow \infty} \left(1 + \frac{-x}{n}\right). \quad (39)$$

Freund ilustration



Estimate Θ by maximum likelihood

The **Method of maximum likelihood** consist in estimate the parameter Θ the number of *red cards* that maximize the probability of see the **data** (three red cards and a blue car), in this case Θ could be 2 or 3.

$$\frac{\binom{3}{2} \binom{1}{1}}{\binom{4}{3}} > \frac{\binom{2}{2} \binom{2}{1}}{\binom{4}{3}} \quad (40)$$

in Scipy the notation is:

$$p(k, M, n, N) = \frac{\binom{n}{k} \binom{M-n}{N-k}}{\binom{M}{N}} \quad (41)$$

```
from scipy.stats import hypergeom
k,M,n,N = 2,4,3,3
print(hypergeom.pmf(k,M,n,N))
k, M,n,N = 2,4,2,3
print(hypergeom.pmf(k,M,n,N))
```

Maximun Likelihood Estimation (MLE)

Suppose that you data it is generated by a theoretical distribution, the inverse problem is determine the most probable parameter that generate the data.

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we are going to say in a general term that $f(x_i)$ is PDF or PMF of a random variable.

Problem

In a formal way we can assume that $x \sim B(n, \Theta)$ and we have seen the data results $\{x_1, x_2, x_3, \dots, x_n\}$

$$P(X = x_1 \cap X = x_2, \dots, \cap X = x_n) \quad (42)$$

The variables are i.i.d and therefore the joint probability is the result of multiply the marginal probabilities.

$$P(X = x_1 \cap X = x_2, \dots, \cap X = x_n) = \prod_{i=1}^n P(x_i) \quad (43)$$

Problem

$$\begin{aligned}\prod_{i=1}^n P(x_i) &= \prod_{i=1}^n \binom{m}{x_i} \Theta^{x_i} (1 - \Theta)^{m-x_i} \\ &= \prod \binom{m}{x_i} \prod \Theta^{x_i} \prod (1 - \Theta)^{m-x_i} \\ &= \prod \binom{m}{x_i} \Theta^{\sum x_i} (1 - \Theta)^{\sum (m-x_i)} \\ \ln(\prod P(x_i)) &= \ln\left(\binom{m}{x_i}\right) + \sum x_i \ln(\Theta) + (nm - \sum x_i) \ln(1 - \Theta)\end{aligned}\tag{44}$$

$$\begin{aligned}
 L(\Theta) &= \ln\left(\prod P(x_i)\right) \\
 \frac{dL(\Theta)}{d\Theta} &= \frac{\sum x_i}{\Theta} - \frac{nm - \sum x_i}{1 - \Theta} = 0 \\
 \frac{1}{\Theta} &= \frac{nm - \sum x_i}{\sum x_i} + 1 \\
 \Theta^* &= \frac{\sum x_i}{nm}
 \end{aligned} \tag{45}$$

Now to proof that $L(\Theta^*)$ is the maximum probability we must show that $\frac{d^2L(\Theta)}{d\Theta^2} \big|_{\Theta=\Theta^*} < 0$.

The weibull distribution is useful for modelling time until an event occur its probability density function is:

$$f(t) = \frac{k}{\lambda} \left(\frac{t}{\lambda}\right)^{k-1} \exp\left(-\frac{t}{\lambda}\right)^k, \quad t \geq 0. \quad (46)$$

where $\lambda > 0$ is scale parameter and $k > 0$ is shape paremeter, when $k = 1$ then weibull is exponential distribution.

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Survival function

